

Miocene regional hotspot-related uplift, exhumation, and extension north of the Snake River Plain: Evidence from apatite (U-Th)/He thermochronology

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ABSTRACT

Passage of North America over the Yellowstone hotspot has had a profound influence on the topography of the northern Rocky Mountains region. One of the most prominent hotspot-related topographic features is the Yellowstone crescent of high terrain, which consists of two elevated shoulders bounding the eastern Snake River Plain and converging at a topographic swell centered on the Yellowstone region. We have applied single-grain (U-Th)/He dating to apatites (AHe) collected from the Pioneer-Boulder Mountains on the northern arm of the Yellowstone crescent of high terrain to constrain the timing, rates, and spatial distribution of exhumation. These data provide constraints on the timing and processes responsible for uplift related to passage of the hotspot. The Pioneer-Boulder Mountains represent a topographic and structural culmination defined by elevation and by the geometry of preserved strata of the Eocene Challis volcanic province. AHe ages indicate that ≥ 2 –3 km of exhumation has occurred in the core of the Pioneer-Boulder Mountains culmination, where no Challis volcanics are preserved, since ca. 11 Ma. Challis volcanics are extensively preserved and Eocene topographic highs are locally preserved to the north and south of the Pioneer-Boulder Mountains, indicating minimal erosion in those areas. Age-elevation relationships suggest an exhumation rate of ~ 0.3 mm/yr between ca. 11 and 8 Ma for the culmination core; this relatively rapid interval of exhumation followed a period of >30 m.y. during which little to no regional-scale exhumation occurred. Spatial patterns of both exhumation and topography indicate that faulting was not the primary control on uplift and exhumation of the culmination. Instead, NNW-trending normal faults are superimposed on the culmination, with the AHe ages from the footwall of the Copper Creek fault indicating that faulting began at or after ca. 10–9 Ma. Regional exhumation at 11–8 Ma was synchronous with silicic eruptions from the ca. 10.3 Ma Picabo volcanic field located immediately to the south and with S tilting of the southern flank of the Pioneer-Boulder Mountains culmination, which was likely the result of loading of the eastern Snake River Plain by midcrustal mafic intrusions. This synchronicity suggests a causal relationship between hotspot processes and exhumation through potential contributions of flexure and mantle dynamics to uplift, as well as changes in drainage networks and base level.

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INTRODUCTION

Although best known for its geothermal features and as the largest active felsic volcanic center in the world, the Yellowstone hotspot has also had profound effects on the topography of the northern Rocky Mountains region. The Yellowstone region is characterized by a topographic swell estimated to be 350–400 km across and up to 1 km in height compared with surrounding regions (e.g., Pierce and Morgan, 1992, 2009; Smith and Braile, 1993; Smith et al., 2009). This elevation anomaly extends from Yellowstone to the southwest as two high-elevation ridges or “arms” that flank the eastern Snake River plain (ESRP, Fig. 1) and was termed the Yellowstone crescent high terrain by Pierce and Morgan

(1992). Each arm of the Yellowstone crescent of high terrain contains a drainage divide that parallels the eastern Snake River Plain, an ~ 90 -km-wide trough largely covered by Pliocene and younger basalts, with the inner sides draining into the Snake River (Fig. 1).

This topographic pattern is the result of southwestward motion of the North American plate over the Yellowstone hotspot, which is indicated by the NE younging of silicic volcanic centers over the last ~ 10 –12 m.y. toward the currently active Yellowstone caldera (Fig. 1; e.g., Suppe et al., 1975; Pierce and Morgan, 1992; Smith and Braile, 1993). Most discussions of the elevated Yellowstone crescent of high terrain attribute uplift to dynamic-thermal uplift above the hotspot (typically a plume tail) as envisioned for the Yellowstone region today (Pierce and Morgan, 1992; Smith and Braile, 1993; Waschbusch and McNutt, 1994; Lowry et al., 2000). Flexural uplift, resulting from loading of the eastern Snake River Plain by mafic flows and emplacement of midcrustal mafic magmas, may

have also contributed to flank uplift, but the little work that has addressed flexure has focused on downflexing (McQuarrie and Rodgers, 1998; Wegmann et al., 2007) rather than potential uplift. The Yellowstone crescent of high terrain has been cut by Basin and Range normal faults, producing additional topography (e.g., Anders et al., 1989; Pierce and Morgan, 1992).

Compared with the large number of studies focused on the spatial-temporal-geochemical evolution of the volcanic rocks, fewer detailed studies have addressed the topographic evolution of the Yellowstone crescent of high terrain and eastern Snake River Plain. Existing work includes studies of drainage evolution adjacent to the ca. 6.7–4.3 Ma Heise volcanic field (Morgan and McIntosh, 2005) and Yellowstone regions (e.g., Fritz and Sears, 1993), detrital zircon provenance (e.g., Beranek et al., 2006; Hodges et al., 2009), fossil mollusc distribution (Taylor and Bright, 1987), and filtered digital elevation model (DEM) analysis (Wegmann et al., 2007). However, there are few data

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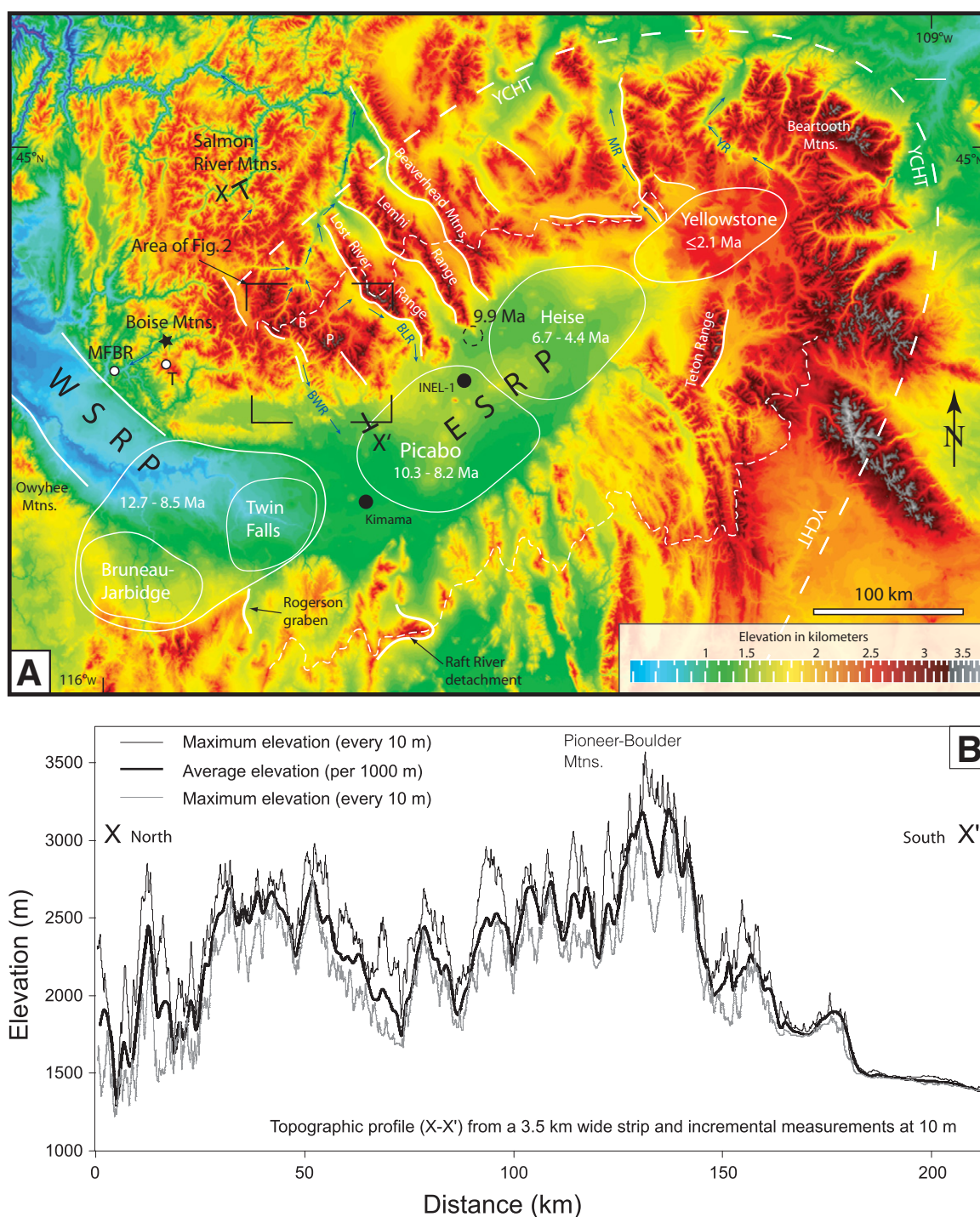


Figure 1. (A) Digital elevation model (DEM) of Snake River Plain and Yellowstone regions. Short dashed lines show position of drainage divides on flanks of eastern Snake River Plain (ESRP). Long dashed lines define outer margin of Yellowstone crescent of high terrain (YCHT). Solid white lines are selected normal faults. Solid white outlines are silicic volcanic centers with ages (Pierce and Morgan, 1992; Bonnichsen et al., 2008). Dashed black circle shows age of 9.9 Ma Howe Point basalt. Kimama and INEL-1 wells are shown as solid black circles. White circles in Boise Mountains are locations of late Miocene AHe ages (Table 1) from Trinity Mountain (T) and the Middle Fork Boise River (MFBR) areas. Black star is location of ca. 11 Ma apatite fission-track (AFT) age of Sweetkind and Blackwell (1989). See Figure 2 for more detailed maps of main study area. P—Pioneer Mountains; B—Boulder Mountains; BWR—Big Wood River; BLR—Big Lost River; WSRP—western Snake River Plain. Blue arrows show flow direction of rivers. **(B)** Topographic profile from eastern Snake River Plain to Salmon River Mountains. Location shown in A. Maximum and minimum elevation profiles were created by taking elevations at 10 m intervals across a 3.5 km width perpendicular to line of section. Average elevation profile (thick black line) was created from 1000 m intervals. MR—Madison River; YS—Yellowstone River.

that directly constrain the timing, amount, and spatial patterns of Miocene and younger uplift and exhumation, particularly in regions adjacent the hotspot at ca. 11–8 Ma (area near the Twin Falls and Picabo volcanic fields; Fig. 1). As a result, we have only a broad understanding of the topographic evolution of the region and the processes responsible for this evolution.

Low-temperature thermochronology provides a record of exhumation-related cooling and, in the absence of tectonic exhumation (or additional heat sources), yields information about the timing, rates, and spatial distribution of erosion. Thus, when combined with other geologic data, low-temperature thermochronology can be used to investigate the interactions among erosional exhumation, uplift, and surficial processes (e.g., Farley, 2002; Ehlers, 2005; Spotila, 2005; Reiners and Brandon, 2006). In this paper, we document the timing and spatial patterns of exhumation derived from apatite (U-Th)/He (AHe) thermochronology from the Pioneer-Boulder Mountains, which lie on the northern arm of the Yellowstone crescent of high terrain (Figs. 1 and 2). These data are used to investigate the topographic evolution of the area and the processes responsible for uplift and exhumation.

The Pioneer-Boulder Mountains represent a broad area of high elevations with numerous peaks and ridges above 3500 m (Figs. 1 and 2B). Existing geophysical data suggest a crustal thickness of ~34–37 km for the Pioneer-Boulder Mountains region (e.g., Yuan et al., 2010; Eager et al., 2011), indicating that the high elevations are not supported by a deep crustal root. Throughout much of the northern U.S. Rocky Mountains, similarly high elevations are found only in Laramide uplifts or as narrow linear ranges in the footwalls of large-scale Miocene normal faults, including the Lost River, Lemhi, and Beaverhead Ranges to the east of the Pioneer-Boulder Mountains (Fig. 1). The broad distribution of elevated topography and the lack of large-scale Basin and Range normal faults bounding the Pioneer-Boulder Mountains suggest that uplift occurred by a mechanism other than footwall uplift.

Vast areas of the Pioneer-Boulder Mountains are capped by Eocene Challis volcanic rocks (e.g., Dover, 1981; Snider, 1995; Worl et al., 1991), indicating minimal erosional exhumation since ca. 46 Ma. However, in the northern Pioneer and southern Boulder Mountains, Challis volcanics have been nearly completely eroded (Fig. 2), leaving only erosional remnants of the basal volcanics and conglomerates on high-elevation peaks and ridges. In a few places, Challis removal coincides with the footwalls of normal faults; however, most of the region of Challis removal is not bounded by normal faults.

With these characteristics, the Pioneer-Boulder Mountains offer an opportunity to investigate hotspot-related uplift and exhumation and to separate effects of normal faulting and hotspot-related uplift. We performed single-grain (U-Th)/He dating of apatite (AHe) samples from the Eocene granitic bodies exposed in the region where Challis volcanics have largely been removed by erosion to test the idea that exhumation and culmination formation in this area occurred in the Miocene, and resulted from uplift related to passage of the Yellowstone hotspot. We document the timing, spatial distribution, amount, and rates of exhumation, and we discuss their implications for the roles of faulting, hotspot-related processes, and surface processes in the exhumation and topographic development for this region of the Yellowstone crescent of high terrain.

YELLOWSTONE HOTSPOT BACKGROUND

A widely espoused, but not universally accepted, model for mid-Miocene and younger volcanism in the northwest United States involves southwest motion of the North American lithosphere (rate of ~25 mm/yr) over a plume that evolved from a large head to a narrower tail (e.g., Pierce and Morgan, 1992, 2009; Smith and Braile, 1993). This model involves: (1) initial impingement of a mantle plume head with the base of the North American lithosphere at ca. 17–16 Ma to produce the Columbia River flood basalts between northern Nevada and eastern Washington and (2) a transition to a narrower plume tail by ca. 10 Ma, leading to a series of NE-younging silicic volcanic centers now buried by younger basalts along the eastern Snake River Plain (Suppe et al., 1975; Richards et al., 1989; Pierce and Morgan, 1992, 2009; Smith and Braile, 1993; Camp, 1995; Smith et al., 2009). Due to the lack of a clear deep mantle plume imaged by seismic tomography and anomalous spatial-temporal patterns of volcanism, nonplume models have also been proposed for the Yellowstone hotspot–Snake River Plain system (e.g., Christiansen et al., 2002; James et al., 2011; Fouch, 2012). In particular, several recent models have suggested that volcanism is driven by mantle convective cells controlled by stalled or sinking oceanic lithospheric slabs (e.g., Sigloch et al., 2008; Faccenna et al., 2010; James et al., 2011).

Southwestward motion of the North American plate over this evolving hotspot system has led to the development of the eastern Snake River Plain, Yellowstone crescent of high terrain, and Yellowstone topographic swell. The low elevations (~1500 m) of the eastern Snake

River Plain have been attributed to loading by emplacement of midcrustal mafic magmas accompanied by lower-crustal flow away from the plain (McQuarrie and Rodgers, 1998; Rodgers et al., 2002) and thermal contraction from cooling following movement away from the hotspot (Brott et al., 1981; Smith and Braile, 1993; Anders and Sleep, 1992). A midcrustal mafic sill network, ~10–12 km thick, along the length of the eastern Snake River Plain is well documented by seismic data and by gravity modeling (Braile et al., 1982; Sparlin et al., 1982; Peng and Humphreys, 1998; DeNosaquo et al., 2009). Some of this mafic input was synchronous with (and the cause of) crustal melting and formation of the NE-younging silicic volcanic centers, while some may have postdated silicic volcanism and provided the source for younger, fractionated basalts that overlie the silicic rocks (e.g., Shervais et al., 2006).

The topographic swell currently centered on the Yellowstone region at the apex of the Yellowstone crescent of high terrain is believed to result from dynamic and thermal buoyancy above the plume tail (Pierce and Morgan, 1992, 2009; Waschbusch and McNutt, 1994; Smith et al., 2009; Lowry et al., 2000). The exact dimensions of the swell, however, are obscured by Laramide basement uplifts, Basin and Range normal faults, and the Eocene Absaroka volcanics.

Evidence for downflexing of the eastern Snake River Plain includes: (1) the southward decrease in elevations of mountain ranges toward the eastern Snake River Plain (Fig. 1), (2) a southward increase in the plunges of Cretaceous fold hinges (McQuarrie and Rodgers, 1998), and (3) southward tilting of Miocene volcanics (Michalek, 2009). Models involving load-induced flexure implies a potential contribution of flexural upwarping (equivalent to a peripheral bulge in foreland basins) to the uplift of the Yellowstone crescent of high terrain and drainage divide that parallels the eastern Snake River Plain. This process, however, has not been adequately investigated.

Given these processes, the high elevations of the arms of the Yellowstone crescent of high terrain may result from (1) dynamic-thermal uplift above the hotspot, as envisioned for the Yellowstone region today, but migrating NE with the hotspot, and/or (2) flexural uplift resulting from loading of the eastern Snake River Plain by midcrustal mafic magmas. In the typical model applied to Yellowstone–eastern Snake River Plain region (from ca. 10 Ma to present), topography develops in three stages: (1) a precaldera phase, where a topographic swell develops over the mantle plume as a result of heating and partial melting, transitioning into (2) a phase of explosive silicic volcanism and caldera collapse

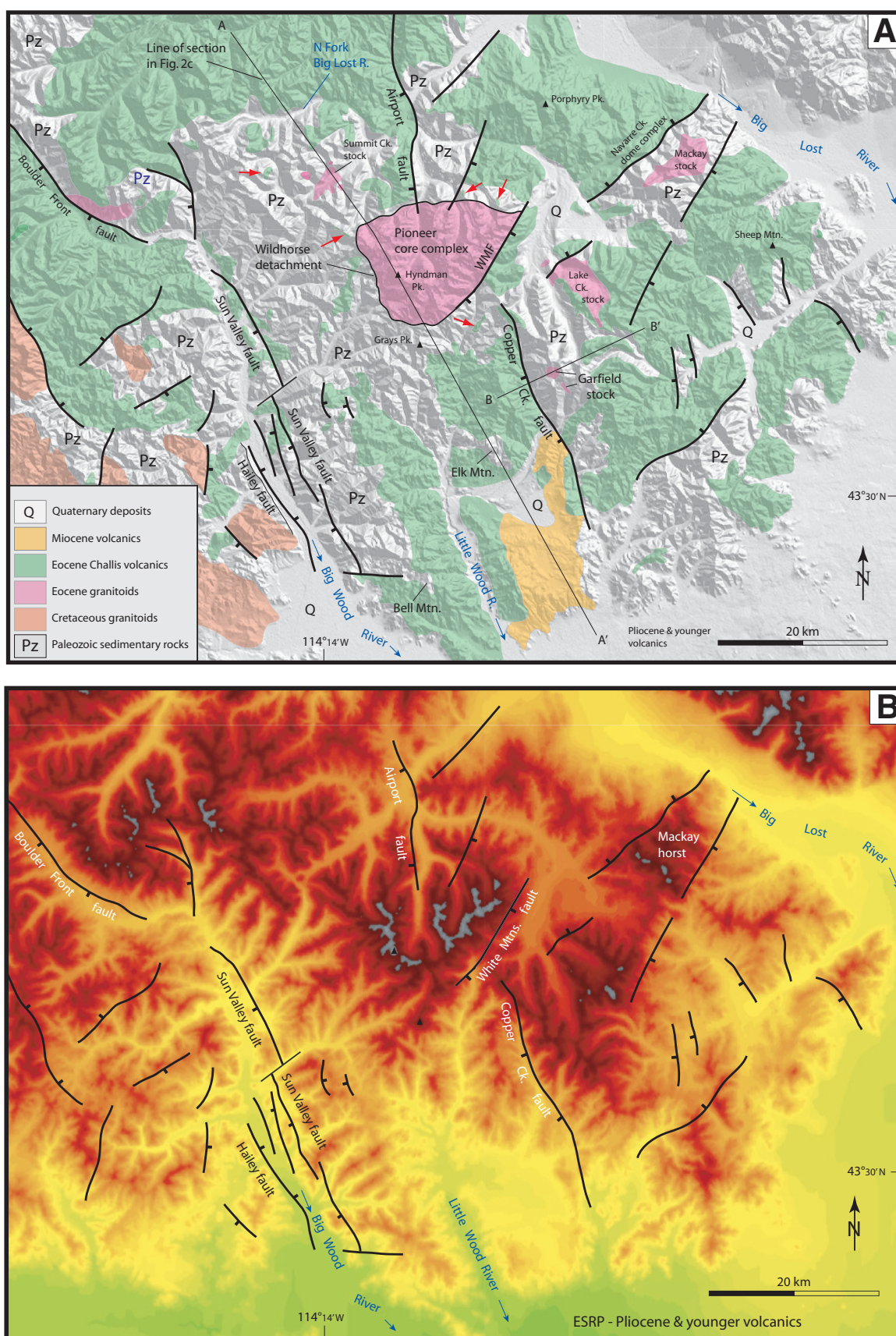


Figure 2 (on this and following page).

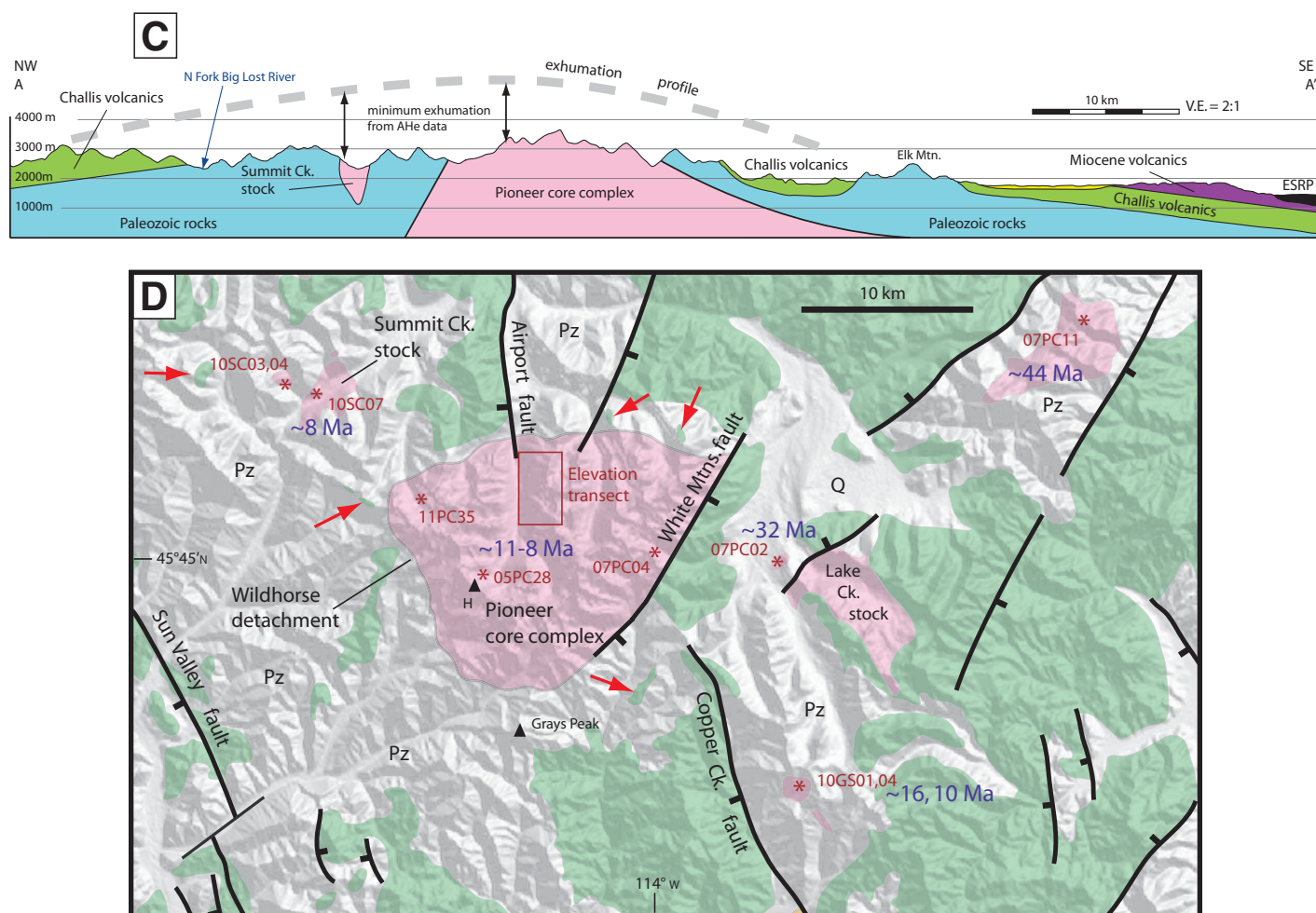


Figure 2 (continued). (A) Geologic map of the Pioneer-Boulder Mountains superimposed on a digital elevation model (DEM) shaded relief map. Red arrows indicate isolated exposures of Challis volcanics located on high-elevation ridges/peaks, but they are too small to be clearly visible on map. These are more visible in part D. Positions of cross sections in part C and Fig. 6B are shown. Geology is modified from Worl et al. (1991), Link and Rodgers (1995), Breckenridge et al. (2003), and Skipp et al. (2009). WMF—White Mountains fault. (B) DEM of area shown in A. Same elevation scale as in Figure 1A. ESRP—eastern Snake River Plain. (C) Geologic cross section across the Pioneer-Boulder Mountains showing the geologic, topographic, and exhumation culmination. Wide dashed gray line is an estimate of amount of exhumation along the section based on AHe data from the culmination core (Fig. 4), preservation of thick Eocene Challis volcanics, and preservation of an Eocene topographic high at Elk Mountain. (D) Detailed map of part of A showing AHe sample locations. The red arrows point out the erosional remnants of Challis volcanics and early Challis conglomerates exposed on high-elevation ridges and peaks discussed in text. Legend is the same as in A.

centered on the topographic swell, and (3) eastern Snake River Plain subsidence caused by thermal contraction, basaltic volcanism at the surface, and loading by emplacement of midcrustal mafic magmas (Pierce and Morgan, 1992; Smith and Braile, 1993), leaving behind the two arms of the Yellowstone crescent of high terrain. Thus, according to this model, uplift followed by subsidence has been a steady-state process, and the topographic swell and uplift rates observed around Yellowstone today can be projected back in time and used to infer the topographic patterns in regions to the SW over the past 10 m.y. (e.g., Pierce and Morgan, 1992, 2009; Smith and

Braile, 1993; Beranek et al., 2006). For example, at ca. 12–10 Ma, the area of the eastern Snake River Plain immediately south of the Pioneer-Boulder Mountains (Fig. 1) could have looked similar to the Yellowstone region today.

Observations of drainage reversal that coincided with formation of the Heise volcanic field are consistent with this model (e.g., Fritz and Sears, 1993). Hodges et al. (2009) showed that ca. 695 Ma detrital zircons derived from the Pioneer core complex in the Pioneer-Boulder Mountains were deposited in the central Snake River Plain by the earliest Pliocene, indicating drainage integration and eastern Snake River

Plain subsidence by that time. However, these observations provide only a minimum age for subsidence. On the basis of fossil mollusc distribution, Taylor and Bright (1987) inferred a drainage divide near the Picabo volcanic center that is constrained only as late Miocene or early Pliocene. Thus, there are few quantitative constraints on the timing and rates of exhumation and uplift in the southwestern half of the eastern Snake River Plain.

Other arguments suggest instead that downflexing of the eastern Snake River Plain occurred early in the volcanic history. Rodgers et al. (2002) provided evidence for significant SE tilt-

ing by ca. 10 Ma as far east as the Beaverhead Mountains (Fig. 1). Furthermore, McCurry and Rodgers (2009) argued from geochemical mass-balance constraints that much of the midcrustal sill inferred from seismic data was the cause of, and therefore was emplaced at the same time as, silicic volcanism. Overall, the age(s) of emplacement of the mafic midcrustal sill is unknown, as is the thickness of basalt that overlies silicic volcanics along the eastern Snake River Plain. Thus, the age of loading remains ambiguous.

The Yellowstone crescent of high terrain has been cut by N- to NNW-trending Miocene faults associated with Basin and Range extension. Several workers have noted that the age of Basin and Range faulting along the southern margin of the eastern Snake River Plain displays a NE-younging trend that is generally coincident with the position of the migrating hotspot (Allmendinger, 1982; Anders et al., 1989; Rodgers et al., 1990), although there are few age constraints on faulting north of the eastern Snake River Plain. Anders et al. (1989) and Anders and Sleep (1992) have attributed this spatial-temporal relationship to increased extensional strain rates brought about by thermal weakening of the lithosphere from intrusion of magmas above the migrating hotspot. They suggested that prior to weakening and following strengthening from cooling/solidification, the region was extending at lower rates.

REGIONAL GEOLOGY AND TOPOGRAPHY

The Pioneer-Boulder Mountains lie on the northern arm of the Yellowstone crescent of high terrain immediately northwest of the ca. 10.3–8.2 Ma Picabo volcanic field (Fig. 1; Pierce and Morgan, 1992), for which the boundaries and post-10.1 Ma eruptive histories are poorly defined (Kellogg et al., 1994). The Pioneer-Boulder Mountains are dominated by Paleozoic sedimentary rocks that were shortened during the Cretaceous Sevier orogeny and that are regionally unconformably overlain by the Eocene (ca. 50–45 Ma) Challis volcanics (e.g., Dover, 1981; Worl et al., 1991; Rodgers et al., 1995). Both the Eocene volcanics and Paleozoic rocks were intruded by Challis-age granitic plutons, which serve as the only lithologies that provide ample, high-quality, large (>100 μm long) apatite for AHe thermochronology. Eocene NW-SE extension produced NE-trending normal faults in the region. Two of the most significant Eocene structures are the Mackay horst and the Pioneer core complex (Fig. 2; e.g., Wust, 1986; O'Neill and Pavlis, 1988; Silverberg, 1990; Vogl et al., 2012). The widespread preservation of Challis volcanics suggests minimal amounts of regional-scale exhumation

between ca. 45 Ma and 11 Ma; however, post-Challis Eocene exhumation occurred in the footwalls of some normal faults during this time. It is important to note that exhumation due to motion of the detachment faults that bound the Pioneer core complex ended in the latest Eocene, such that any exhumation that occurred during passage of the Yellowstone hotspot is unrelated to the core complex structures.

The Pioneer-Boulder Mountains contain the highest elevations within a NW-SE transect across the Yellowstone crescent of high terrain at $\sim 114^\circ\text{W}$ longitude and contain numerous peaks and ridgelines at elevations around 3500 m. Elevations decrease southward to ~ 1400 –1500 m in the eastern Snake River Plain (Fig. 1B). North of the Pioneer-Boulder Mountains, elevations decrease more gradually toward the Salmon River Mountains (Fig. 1). The Pioneer-Boulder Mountains differ structurally and topographically from the Lost River, Lemhi, and Beaverhead Ranges to the east in that (1) the Pioneer-Boulder Mountains are not bounded by large-displacement, laterally continuous normal faults, and (2) the highest elevations in the Pioneer-Boulder Mountains are more distributed compared with the long, narrow topographic highs found in the immediate footwalls of the ranges to the east.

The regional drainage divide separating streams that drain into the eastern Snake River Plain from those that drain to the north (and eventually west) into the Salmon River system occurs within the Boulder Mountains (Fig. 1). This divide continues to the east, transecting the ranges and valleys created by the large-scale Miocene normal faults. Higher-order drainage divides in the Pioneer-Boulder Mountains are controlled by lithology, as well as by Miocene and Eocene faults.

Intermediate to felsic lava and pyroclastic flows of the Challis volcanic group erupted across much of central Idaho during the period of ca. 50–45 Ma (Moye et al., 1988; Snider, 1995; Sanford, 2005). These volcanics may have filled much of the preexisting topography that formed during Laramide-age thrusting and early phases of Eocene extension (Janecke, 1992; Snider, 1995). At least 1–2 km of volcanics accumulated in some areas, with the greatest thicknesses occurring near eruptive centers (e.g., Snider, 1995). It is difficult, however, to reconstruct the regional thickness variations in detail due to erosion and lack of exposure of the basal contacts. Because Challis volcanics erupted over a large region, the extent of Challis removal or preservation serves as a first-order indicator of the magnitude of exhumation, and the elevation of the base of Challis units serves as a marker of the relative uplift of different

areas. However, in detail, the base of the Challis volcanics was affected by preexisting topography and small-scale faults (throws generally $\ll 500$ m).

In the southern Boulder and northern Pioneer Mountains, the Challis volcanics (or basal Challis conglomerate) are completely eroded or occur only as erosional remnants on peaks and ridges (Fig. 2). In contrast, Challis volcanics are extensively preserved to both the north and south (e.g., Worl et al., 1991). Thus, the distribution of Challis volcanics across a NNW-SSE transect defines a structural high defined by the elevation of the base of the Challis volcanics and an exhumation culmination where Challis volcanics were extensively removed (Fig. 2C). This culmination broadly coincides with the topographic culmination in the northern Pioneer and Boulder Mountains described earlier herein. The presence of early Challis conglomerates as isolated remnants on high-elevation peaks and ridges (Fig. 2A) indicates that Eocene topography has been inverted and completely removed within the core of the culmination. This culmination is discussed in more detail in a later section in conjunction with the AHe data.

The removal of the Challis volcanics has exposed the underlying Paleozoic sedimentary rocks, which were intruded by a number of Eocene granitic bodies (Fig. 2A) emplaced at depths ranging from subvolcanic to 11–15 km (Silverberg, 1990; Vogl et al., 2012). Partial exhumation from Eocene extensional faulting brought the deeper plutons to upper-crustal depths by the end of the Eocene. Figure 3, which is constructed from geologic observations and the AHe data in this study, shows the estimated positions of each of the sampled granitic bodies with respect to the AHe partial retention zone (PRZ), or depth associated with the temperature range of 40–80 $^\circ\text{C}$, immediately prior to the late Miocene. This figure and the following discussions show that despite the variation in emplacement depths, all of the granitic bodies appear to have resided within the uppermost 3–4 km by the middle Miocene, but the different sample locations underwent varying amounts of Miocene and younger exhumation. Our AHe dating from these bodies allows quantification of the age, amount, and rates of exhumation within the culmination.

(U-Th)/He THERMOCHRONOLOGY BACKGROUND, SAMPLING, AND METHODS

The (U-Th)/He thermochronology method is based on the production of alpha particles (^4He atoms) from radioactive decay of ^{238}U , ^{235}U , ^{232}Th , and ^{147}Sm (Farley, 2002). The ingrown ^4He

atoms experience temperature-driven diffusion for which the rate is primarily determined by the type of host materials. The diffusion of helium in apatite is relatively sensitive to temperatures (Zeitler et al., 1987; Wolf et al., 1996), causing essentially no accumulation of ingrown ^4He above $\sim 80^\circ\text{C}$ for most apatites in natural environments. For a typical grain radius of 70–180 μm and a cooling rate of $\sim 10^\circ\text{C}/\text{m.y.}$, the AHe closure temperature is estimated as ~ 70 – 80°C based on stepped heating diffusion experiments (Farley, 2000; Shuster et al., 2004). However, even below the “closure temperature,” slower He diffusion continues, allowing only partial retention of ingrown ^4He in apatite. This slower diffusion, or partial He retention in apatite, continues until the temperature decreases to $\sim 40^\circ\text{C}$, at which point the He diffusion effectively ceases. The vertical segment corresponding to the temperature range of ~ 80 – 40°C is called the “partial retention zone” (PRZ; Fig. 3; Wolf et al., 1996, 1998; Stockli et al., 2000; Farley, 2002). Because the AHe closure temperature is lower than any other established thermochronologic method, the AHe system provides a means with which to document the lowest-temperature thermal history of the upper crust (2–3 km or less), and, in most cases, it records information about the transit of rocks through the shallowest crustal levels.

Samples for AHe dating were collected from Eocene granitic rocks (Fig. 2D). Samples were collected through the maximum elevation range of exposure for each body. However, many of the bodies are exposed over only a few hundred meters of elevation; our maximum elevation profile is from the central Pioneer Mountains (Fig. 2D), where granitic rocks are exposed over >1100 m of elevation.

Most of the (U-Th)/He analyses were undertaken at the University of Florida, except five samples (05PC28, 07PC01, 07PC02, 07PC04, 07PC16) for which ^4He measurements were

performed at the University of Arizona. Outcrop samples were processed following standard mineral separation procedures of crushing, sieving, magnetic separation, and heavy liquid methods. To avoid apatite grains with inclusions, the grains were carefully examined under a stereomicroscope with open and cross-polarized conditions at maximum magnification of $\times 160$. Apatite grains with visible inclusions were discarded. Selected grains were examined using a scanning electron microscope (SEM) to confirm their mineral identity/purity and to examine basic morphological features. To minimize artificial modifications of the apatite grains during the SEM analysis, the individual samples were placed on standard scotch tape without a coating procedure, and then analyzed at a variable pressure mode. Shan et al. (2013) showed that extended chemical mapping up to ~ 2 h, even at high SEM beam currents (1000 pA), does not cause any detectable modification of (U-Th)/He ages for the Durango apatite standard. After SEM analysis, all the grains were examined under a stereomicroscope to determine their physical dimensions. Grains that were selected for analysis were between 80 and 250 μm in length.

At the University of Florida, single grains of apatite were wrapped in Nb or Pt packets and placed in an ~ 3 -mm-deep well within a stainless-steel planchette. The loaded planchette was sealed in a high vacuum, and the individual packets were heated using a diode laser at 7 amps for 3 min. The extracted gas was mixed with a known amount of $>99.99\%$ pure ^3He spike, purified with a NP-10 getter, and then analyzed with a Pfeiffer-Balzars Prisma quadrupole mass spectrometer. All the grains were re-extracted twice to confirm complete He degassing of samples. Most of the samples yielded negligible amounts of gas after the first extractions. For all samples, the second re-extraction contributed

less than 0.5% of the total measured ^4He . Procedural blanks were measured after every three samples. Durango apatite standards were measured every 10 samples. The entire He analysis sequence was performed in an automatic mode using Labview (National Instruments) and auto-click (MurGee) codes. The degassed sample packets were retrieved from the sample chamber, transferred to Teflon vials, mixed with ~ 0.05 mL of U-Th-Sm spike, and then dissolved in 5% nitric acid at $\sim 120^\circ\text{C}$ for inductively coupled plasma–mass spectrometry (ICP-MS) analysis. The abundances of U, Th, and Sm were determined using a Thermo-Finnigan Element2 ICP-MS. Alpha ejection effects were corrected based on the physical dimensions of the grains (Farley et al., 1996). Analytical errors for the (U-Th)/He ages were propagated using the Monte-Carlo simulation. An additional 3% of α -recoil correction errors at 1σ level were included in the error calculation.

AHe DATING: RESULTS AND INTERPRETATION OF EXHUMATION PARAMETERS

Pioneer Core Complex and Summit Creek Stock Areas

Background

The Pioneer core complex is bounded by the curvilinear Wildhorse detachment and by the White Mountains fault on its southeast side (Fig. 2A). The footwall consists of Ordovician to Archean orthogneisses and paragneisses that were intruded by granitic rocks at ca. 50–47 Ma (Vogl et al., 2012). The $^{40}\text{Ar}/^{39}\text{Ar}$ analyses of muscovite and biotite yield ages between ca. 38 and 35 Ma (Silverberg, 1990). The low-temperature steps in K-feldspar record $^{40}\text{Ar}/^{39}\text{Ar}$ ages of ca. 33 Ma (Silverberg, 1990), suggesting that cooling below 175–200 $^\circ\text{C}$ occurred by the end of the Eocene.

The main body and smaller satellite bodies of the Summit Creek Stock are exposed ~ 7 – 9 km northwest of, and in the hanging wall of the Pioneer core complex (Fig. 2A). The main body of the Summit Creek Stock is texturally and compositionally similar to the 48–47 Ma granitoids in the eastern half of the Pioneer core complex footwall and has been used as a possible piercing point for displacement estimates on the Wildhorse detachment (e.g., Wust, 1986). No U-Pb ages have been published for the Summit Creek Stock, but ages for biotite and K-feldspar low-temperature steps are ca. 45 and 44 Ma, respectively (Silverberg, 1990). These data suggest that the Summit Creek Stock was intruded into relatively shallow ($<300^\circ\text{C}$) depths above the Pioneer core complex at ca. 48–47 Ma and

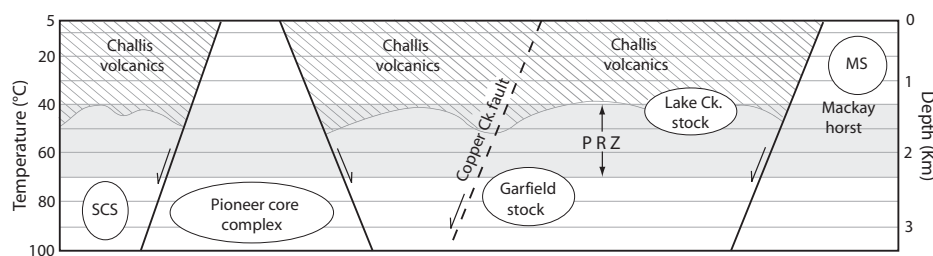


Figure 3. Estimated positions of each of the sampled granitic bodies with respect to the AHe partial retention zone (PRZ, gray shading) immediately prior to late Miocene exhumation event. Solid lines with arrows are Eocene normal faults that produced partial exhumation of Pioneer core complex and Mackay Stock (MS) prior to the Miocene. SCS—Summit Creek Stock. Dashed line is the Copper Creek fault, which was not active prior to the late Miocene. Eocene Challis volcanics are shown in diagonally ruled pattern. Variable and unknown depth to the base of the Challis volcanics is shown schematically. Depth on right vertical axis corresponds to a thermal gradient of $30^\circ\text{C}/\text{km}$ and assumes a mean surface temperature of 5°C .

was translated downward to the west-northwest along the Wildhorse detachment, with displacement continuing through the end of the Eocene (based on differences in the K-feldspar $^{40}\text{Ar}/^{39}\text{Ar}$ cooling ages).

Data

The Pioneer core complex contains the largest sampling region of Eocene granitic rocks in terms of both areal and vertical extent. We analyzed one to three single apatite grains from nine samples collected at elevations between ~2350 m and ~3440 m on a ridge in the north-central part of the core complex (Fig. 2D). Except for three outliers, all of the individual apatite grains from the nine samples yielded ages in the range of 11.9–8.7 Ma, with one slightly older age of 12.4 Ma (Table 1; Fig. 4, black circles). The reason for the three outliers of 5.3, 14.9, and 20.7 Ma is unclear; such scatter can be potentially explained by (1) unsupported ^4He from U-Th-rich phases either inside of or nearby apatite grains (Ehlers and Farley, 2003; Spiegel et al., 2009), or (2) incorrect a-ejection (F_T) correc-

tion due to heterogeneous U-Th distribution in apatites (Hourigan et al., 2005; Ault and Flowers, 2012). These outliers are not considered in interpretations of exhumation histories.

Three additional samples from other areas of the Pioneer core complex were analyzed to assess lateral differences in ages (Fig. 4, gray circles). These samples from southeast, south-southwest, and west of the main elevation transect area (Fig. 2D) yielded single-grain ages between 7.7 and 9.8 Ma, which overlap with the other Pioneer core complex sample ages.

We dated three samples from the Summit Creek Stock bodies (Fig. 2B). Apatites from the lowest-elevation (~2325 m) sample yielded three ages between 7.3 and 8.2 Ma with a weighted-mean age of ca. 7.7 ± 1.3 Ma (Fig. 4, black triangles), although two other grains yielded ages of ca. 2.7 and 17.5 Ma. Six apatites from two other samples from higher elevations have higher intra-sample age variations, yielding ages between ca. 5.2 and 13.6 Ma (Table 1; Fig. 4, open triangles).

Interpretation

Several lines of evidence indicate that the Miocene exhumation recorded by our AHe ages was not the result of motion on the Wildhorse detachment and that the Pioneer core complex and Summit Creek Stock were being exhumed as a single block. (1) Both the Pioneer core complex and Summit Creek Stock yield late Miocene ages, with the youngest age coming from the Summit Creek Stock, which was in the hanging wall of the detachment. (2) Cooling ages discussed here suggest that the Pioneer core complex and Summit Creek Stock were juxtaposed at similar depths by the earliest Oligocene. (3) There is no geomorphic expression of the Wildhorse detachment, which stands in stark contrast to most Miocene faults in the region. (4) The distinctly curvilinear geometry of the Wildhorse detachment formed during NW-SE extension is unlikely to have been able to accommodate SW-NE Basin and Range extension, and no younger faults cut the detachment. (5) Challis volcanics have been removed from above the Summit Creek Stock and in the region between the Summit Creek Stock and Wildhorse detachment (with preservation of basal conglomerates on high ridgelines).

Single-grain AHe ages for both the Pioneer core complex and Summit Creek Stock areas are shown as circles on an age versus elevation plot in Figure 4. Overall, the data in Figure 4 show a slight decrease in age toward lower elevations, without the three outlier ages from 2350 m (10PK12) and ~3078 m (09PC36). Excluding these samples, a linear regression yields an apparent exhumation rate of 0.31 (+0.03/–0.08) mm/yr (Fig. 4). The best-constrained ages from

the Summit Creek Stock (7.3, 7.6, and 8.3 Ma for 10SC07) are consistent with the linear trend of the Pioneer core complex in the age-elevation plot, adding further support to the geologic observations (discussed here) indicating that the Summit Creek Stock and Pioneer core complex areas were exhumed as a coherent block during the Miocene.

Several studies have shown that He retentivity and therefore closure temperatures are affected by radiation damage, and since radiation damage should be proportional to effective U concentration (eU, where $\text{eU} = \text{U} + 0.235\text{Th}$), AHe age may positively correlate with eU, particularly for slowly cooled samples (e.g., Shuster et al., 2006; Flowers et al., 2007, 2009). Ages from the Pioneer core complex and Summit Creek Stock show no correlation with U concentration (Fig. 5), consistent with relatively fast cooling as indicated by the age-elevation relationships in Figure 4.

The exhumation rate derived from the age-elevation trend may be an overestimate of the exhumation rate if the near-surface isotherms are strongly affected by topography and heat advection during rapid exhumation, particularly for low-temperature thermochronometers such as AHe (e.g., Stüwe et al., 1994; Mancktelow and Grasemann, 1997; Braun, 2002). To quantitatively assess the topographic warping effect of the near-surface isotherms, a parameter called “admittance ratio (α)” was introduced (Braun, 2002), which represents a ratio between the relief of an isotherm and the topographic relief. The admittance ratio is zero when the isotherm is flat and not disturbed by a short-wavelength topography. For long-wavelength topography, where the isotherms tend to follow the topography, the admittance ratio is in the range of 0.5–0.8 (Reiners et al., 2003). Our samples were collected from an area with short-wavelength (wavelength ~6 km) topography having a short horizontal sampling distance of only 1.6 km. The admittance ratio is expected to be small for these

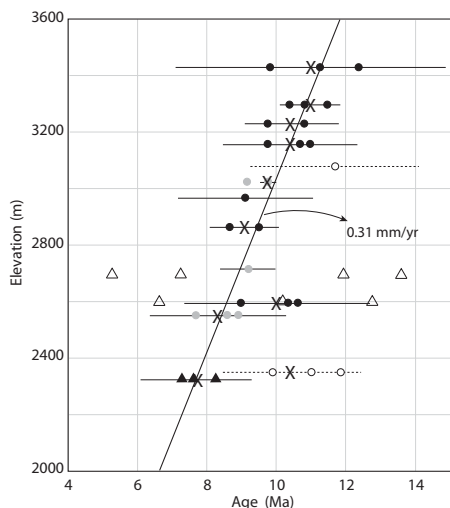


Figure 4. Age versus elevation plot for AHe samples from the Pioneer core complex and Eocene Summit Creek Stock (see Fig. 2A). Black filled circles are single-grain ages for the elevation transect within the Pioneer core complex, and gray circles are single-grain ages for three other samples from different parts of the Pioneer core complex. Black filled triangles are single-grain ages from the Summit Creek Stock. Weighted mean ages for the various samples are indicated with an “X,” with the 2σ uncertainties shown by the horizontal lines. Open symbols with dashed uncertainty lines are samples not used in the linear regression. Best-fit linear regression of ~0.31 mm/yr (excluding the samples shown with open symbols) is shown with black line. Weighted means and regression were calculated using the Isoplot program of Ludwig, 2008.

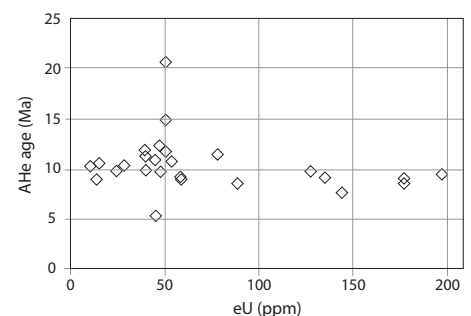


Figure 5. Plot of AHe age versus effective U ($\text{eU} = \text{U} + 0.235\text{Th}$) concentration for Pioneer core complex and Summit Creek Stock samples.

parameters, as well as for the moderate exhumation rate. Assuming a steady-state thermal distribution in two dimensions, these topographic data suggest an admittance ratio of <0.1 for the estimated closure isotherm of $60\text{--}70^\circ\text{C}$, indicating that the isotherm warping due to topography is very limited. Therefore, we conclude that the exhumation rate derived from Figure 4 should closely approximate the true exhumation rate.

The steep age-elevation trend indicates that the current levels of exposures in the Pioneer core complex–Summit Creek Stock areas were located beneath the AHe PRZ until ca. 11 Ma. The AHe closure temperature for the Pioneer core complex–Summit Creek Stock samples is estimated as $60\text{--}70^\circ\text{C}$ based on the grain dimensions ($40\text{--}90\text{ }\mu\text{m}$ in half-width) and approximate cooling rate of $\sim 10^\circ\text{C/m.y.}$ Assuming a two-dimensional steady-state thermal distribution, these mean surface temperature of $\sim 5^\circ\text{C}$ and a high (but reasonable, given the heat-flow values) geothermal gradient of 30°C/km , the data indicate that at least $\sim 2.2\text{ km}$ of material were removed from the ridges since ca. 11 Ma, and over 3.2 km of exhumation occurred in the valleys during the past ca. 7–8 m.y.

Extrapolation of the rate of $\sim 0.3\text{ mm/yr}$ linear best fit in Figure 4 indicates that apatites with ages of zero (i.e., diffusive He loss is occurring at a rate equal to He production) would be found at an elevation of $\sim 100\text{ m}$ below sea level. In contrast, by assuming a mean elevation of $\sim 2600\text{ m}$ with a mean surface temperature of 5°C and a thermal gradient of $\sim 30^\circ\text{C/km}$, temperatures of $\sim 75^\circ$ would be expected at $\sim 600\text{ m}$ above sea level. This may indicate that 11–8 Ma represents a time of maximum exhumation rate and that rates decreased sometime after ca. 8 Ma. However, given the uncertainties in the regression (and thermal gradient over time), we regard this as a tenuous conclusion.

The data plotted in Figure 4 do not show a distinct slope break to define the base of the PRZ, and therefore the age of onset of exhumation is unclear. We note that exhumation beginning prior to ca. 11 Ma would increase the amount of total exhumation. However, since Challis volcanics are preserved along some high ridges between the Pioneer core complex and Summit Creek Stock, a larger total amount of exhumation would require an anomalously thick section of Challis volcanics.

In summary, the Pioneer core complex–Summit Creek Stock areas underwent $2.2\text{--}3.2\text{ km}$ of exhumation since ca. 11 Ma. Given the observed scatter in our data, our best estimate of exhumation rates between ca. 11 and 8 Ma is $\sim 0.3\text{ mm/yr}$. We note that the exact rate of exhumation is not critical to our discussions, and any rate in the range of tenths of a millimeter per

year represents a significant increase in regional exhumation rates from the preceding 35 m.y., as evidenced by the widespread preservation of 50–45 Ma volcanics discussed earlier.

Copper Creek Fault Footwall (Garfield Stock)

The Copper Creek fault (Figs. 2 and 6; Breckenridge et al., 2003; Skipp et al., 2009) is a NNW-trending, W-side-down normal fault with $\sim 2\text{--}4\text{ km}$ of throw (Fig. 6) that loses displacement and geomorphic expression to the north, before reaching the Pioneer core complex. The presence and attitude of Challis volcanics in the eastern footwall suggest that the footwall is tilted gently eastward. We analyzed two samples (three grains from each sample) from the Garfield Stock, which is exposed in the footwall. The highest-elevation sample yielded a weighted mean age of $16.6 \pm 2.1\text{ Ma}$ (Table 1), and a sample from $\sim 240\text{ m}$ lower elevation yielded a weighted mean age of $9.6 \pm 1.8\text{ Ma}$ (Table 1). Another sample from lower elevation did not yield inclusion-free apatite, precluding reconstruction of a more complete age-elevation profile.

We consider the age difference between the two samples to be real, since there is a small intrasample age variation and because the negative correlation between age and eU precludes an explanation by radiation-damage-induced differences in retentivity. The gentle slope defined by the two samples on an age-elevation plot may be interpreted as extended residence in the AHe PRZ. The slope is comparable to samples held isothermally within the PRZ for 40–50 m.y., before being exhumed rapidly through the PRZ (e.g., Wolf et al., 1998; Stockli et al., 2000). Such an extended duration of a stable geotherm/PRZ and long residence of the sample near the base of the PRZ are supported by the preservation of ca. 46–47 Ma Challis volcanics in the eastern footwall, which likely covered the entire footwall prior to faulting and eastward tilting (Fig. 6B). Thus, with the lack of lower-elevation samples, we interpret these elevations as representing the lowermost part of the AHe PRZ (Fig. 3). This requires that exhumation rates must have increased dramatically beginning at or after ca. 9.6 Ma, and it suggests that more than 2 km of exhumation occurred within the footwall of the Copper Creek fault since the inception of the Copper Creek fault at that time.

Lake Creek Stock

We dated two apatite grains from a sample (elevation of $\sim 2550\text{ m}$) of small satellite intrusions of the Lake Creek Stock that are $\sim 6\text{ km}$

ESE of the White Mountains fault, which bounds the southeast side of the Pioneer core complex. The southeastern two-thirds section of the Lake Creek Stock intrudes the Challis volcanics, indicating that the stock is a high-level intrusion, and, according to the map of Skipp et al. (2009), the satellite bodies lie in the hanging wall of a NW-side-down, NE-trending normal fault that separates the satellites from the main body.

The two apatites yielded AHe ages of $29.6 \pm 0.9\text{ Ma}$ and $34.0 \pm 1.1\text{ Ma}$, i.e., intermediate between the ages from the Mackay horst and the Pioneer core complex–Summit Creek Stock region. These ages, combined with the field evidence for shallow pluton emplacement, suggest an extended residence in the upper 1–2 km of the crust, probably in the upper AHe PRZ (Fig. 3).

Mackay Horst and Stock

The Mackay horst is a NE-trending structural and topographic high to the east of the Pioneer core complex. Paleozoic rocks of the horst are intruded by the ca. 48–47 Ma (Snider, 1995) composite Mackay Stock. Rhyolite domes and dikes (Navarre Creek dome complex) in the hanging wall, which are likely the shallow-level equivalents of the Mackay Stock, are exposed in the hanging wall of the northwestern horst-bounding fault (Fig. 2A), indicating a horizontal component of motion of $\sim 6\text{ km}$ (Snider, 1995). Thus, reconstruction across the $40^\circ\text{--}45^\circ$ -dipping fault suggests that the present levels of exposure of the Mackay Stock were likely emplaced at $\sim 5\text{--}6\text{ km}$ depth.

We have dated one sample from the crest of the horst at an elevation of $\sim 3100\text{ m}$. Three apatites from this sample yield a weighted mean age of $44.0 \pm 6.6\text{ Ma}$ (Table 1). The small difference between the emplacement age and AHe age, and the reconstruction discussed earlier herein suggest that most of the motion on the northern horst-bounding fault occurred by ca. 44 Ma and that this fault exhumed the sample to within 1–1.5 km of the surface (i.e., above the AHe PRZ; Fig. 3) by this time. This depth at ca. 44 Ma provides an upper bound for Miocene exhumation.

Boise Mountains

The Boise Mountains on the northeast side of the western Snake River Plain are underlain by Cretaceous granitoids of the Idaho batholith that were locally intruded by Eocene granitoids. These rocks are cut by largely NW-trending normal faults with poorly constrained amounts of displacement.

We have dated two samples from the Boise Mountains to the west of the Pioneer-Boulder

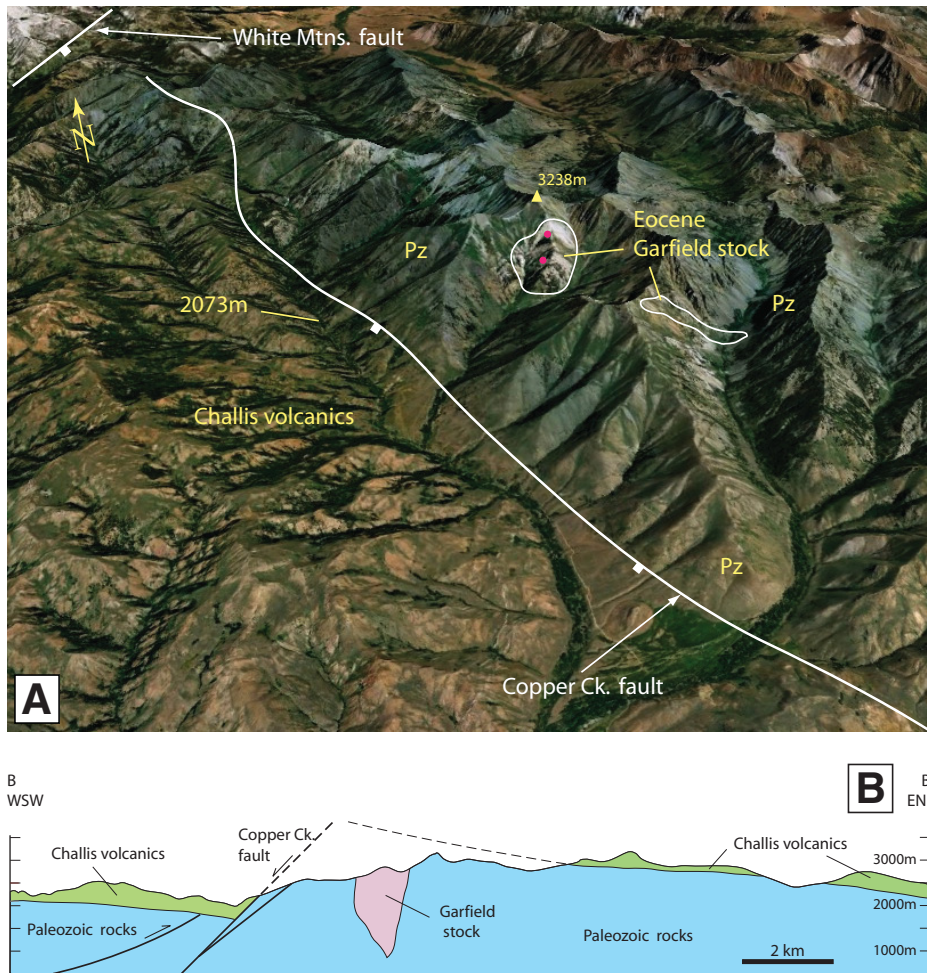


Figure 6. (A) Oblique-view GoogleEarth image of Copper Creek fault. View is approximately toward the north. Total fault length shown is ~18 km. Elevations at peak in footwall of Copper Creek fault and adjacent valley in hanging wall are indicated. Pz—Paleozoic sedimentary rocks. Pink dots are locations of two samples of Garfield Stock discussed in text and provided in Table 1. Note that the geomorphic expression of the fault footwall dies out to the north before reaching the White Mountains fault, which bounds the southeast side of the Pioneer core complex. (B) Geologic cross section across the Copper Creek fault, constructed from the geologic map of Skipp et al. (2009). See Figure 2A for location.

Mountains. Our lowest-elevation sample (~933 m), collected from the Middle Fork Boise River, yielded two ages of 9.5 Ma and 10.1 Ma. The second sample (11TM01) was collected from an elevation of ~2317 m on the slopes of Trinity Mountain, one of the highest peaks (2881 m) in the Boise Mountains. Three single-grain apatite ages from this sample yielded tightly clustered ages of 10.6, 11.1, and 11.2 Ma, with a mean age of 11.0 Ma.

These data suggest that the highest elevations of at least some of the fault blocks of the Boise Mountains underwent at least 2 km of exhumation since ca. 11 Ma. Sweetkind and Blackwell (1989) also presented an apatite fission-track age of ca. 11.4 Ma from the valley floor (~1200 m) along the Middle Fork of the Boise River (star in

Fig. 1A), suggesting that some of the lowest elevations in the Boise Mountains were exhumed from depths beneath the apatite fission-track partial annealing zone.

EXHUMATION AND TOPOGRAPHIC PATTERNS IN THE BOULDER-PIONEER MOUNTAINS

Although the AHe data discussed here come from isolated granitic bodies emplaced at widely varying depths (~1–2 to 11–14 km) in the Eocene, some of these bodies (Pioneer core complex and Mackay Stock) were partially exhumed during postemplacement Eocene extension. On the basis of the AHe data discussed for individual areas herein, in Figure 3

we have provided a summary of depths of the individual bodies with respect to the AHe PRZ immediately prior to Miocene (ca. 11 Ma) exhumation. This figure provides a generalized overview of the different amounts of post-ca. 11 Ma exhumation undergone by each area.

Before discussing controls on exhumation, we first discuss our AHe data together with other geologic observations to fully define the spatial distribution of exhumation. In particular, the extent of preservation of the Challis volcanics and the elevation of the basal Challis contact (Fig. 2) provide a means of defining the uplift and exhumation patterns.

Prior to Miocene erosional exhumation, the Summit Creek Stock area and the region surrounding the Pioneer core complex were likely covered by Challis volcanics of variable thickness, as there are few mapped faults in the region surrounding the Pioneer core complex that would indicate extensive tectonic thinning. Furthermore, Challis volcanics occur locally on peaks/ridges at high elevations (for example, basal Challis conglomerate is preserved on a ridge at an elevation >3100 m immediately north of the Pioneer core complex; Figs. 2A and 2D). The Pioneer core complex and Summit Creek Stock localities that record 2–3 km of post-11 Ma exhumation occur in a region that coincides with nearly complete removal of Challis volcanics, with local preservation only on high-elevation ridges. This area of Challis removal occurs over a NW-SE distance of 30–35 km (Fig. 2C).

North of the Pioneer core complex–Summit Creek Stock area, the North Fork Big Lost River drainage (Boulder Mountains) marks the boundary between areas where Challis deposits are widely preserved to the north (Worl et al., 1991) and the extensively exhumed Pioneer core complex–Summit Creek Stock area to the south (Fisher et al. (1992) and Janecke et al. (1997). Within this drainage, the base of the Challis volcanics occurs at elevations of ~2400–2600 m (Fig. 2) and at elevations <2000 m farther north.

Immediately south of the Pioneer core complex, Challis volcanic and conglomerate units are absent from peaks as high as ~3200 m (Grays Peak), but they do occur as erosional remnants on ridges to the east (Figs. 2A and 2D). Farther south, Challis volcanics are widely preserved and occur at progressively lower elevations (Fig. 2). Minimal erosion south of the Pioneer core complex area is also evidenced by the preservation of topographic highs that existed during Eocene eruption of the Challis volcanics. One example is the Elk Mountain area, where Challis volcanics at elevations <1800 m (base not exposed) surround and onlap peaks composed of Paleozoic sedimentary rocks (Fig. 2). Similar

TABLE 1. (U-Th)/He APATITE DATA

Sample	Elevation (m)	U (ppm)	Th (ppm)	Sm (ppm)	eU (ppm)	⁴ He (fmol)	Mwar (μm)*	F _T	(U-Th)/He age (Ma)	1 [†]	Mean	Latitude (°N)	Longitude (°W)
Pioneer core complex—elevation transect													
10PC39 (1)	3430	30	70	116	47	54.0	97.2	0.85	12.4	0.70	11.14	43°47.930'	114°04.011'
10PC39 (2)		24	62	98	38	31.1	89.7	0.84	11.2	0.65			
10PC39 (3)		16	36	44	24	7.08	60.3	0.78	9.8	0.59			
09PC33 (1)	3298	18	42	40	28	7.30	52.5	0.76	10.4	0.66	10.88	43°48.487'	114°04.481'
09PC33 (2)		34	84	78	54	25.4	64.5	0.80	10.8	0.79			
09PC33 (3)		49	121	95	77	14.0	42.3	0.71	11.5	0.54			
09PC34 (1)	3231	30	72	111	47	42.7	88.5	0.85	9.7	0.89	10.28	43°48.568'	114°04.339'
09PC34 (2)		29	64	92	44	17.4	70.8	0.82	10.8	0.75			
09PC35 (1)	3158	31	63	74	46	7.08	52.5	0.76	9.8	0.52	10.46	43°48.580'	114°04.292'
09PC35 (2)		30	62	67	45	5.84	42.0	0.71	11.0	0.60			
09PC35 (3)		34	80	82	53	5.59	40.0	0.70	10.7	0.63			
09PC36 (1)	3078	33	76	94	51	15.7	53.5	0.77	11.7	2.00	15.79	43°48.658'	114°04.201'
09PC36 (2)		33	74	86	50	30.9	51.5	0.77	20.7 [§]	3.10			
09PC36 (3)		26	99	143	50	51.2	68.8	0.82	14.9 [§]	2.30			
07PC16 (1)	2966	38	27	269	45	5.29	61.2	0.78	5.3 [§]	0.36	7.22	43°47.665'	114°04.644'
07PC16 (2)		157	87	1170	177	14.6	42.0	0.70	9.1	0.81			
10PC21 (1)	2865	155	94	332	177	23.5	44.3	0.73	8.7	0.59	9.08	43°47.966	114°04.893
10PC21 (2)		172	107	298	197	12.7	40.0	0.69	9.5	0.59			
09PC19 (1)	2596	12	13	72	15	0.91	34.3	0.65	10.6	0.76	9.96	43°47.947'	114°05.225
09PC19 (2)		10	2.9	66	10	2.25	60.8	0.79	10.3	0.90			
09PC19 (3)		13	2.6	95	14	2.39	55.3	0.77	9.0	0.88			
10PK12 (1)	2350	27	53	68	39	22.0	79.9	0.82	11.8	2.00	10.91	43°48.561'	114°05.450'
10PK12 (2)		25	59	73	39	9.06	56.2	0.76	9.9	0.56			
10PK12 (3)		26	62	84	40	24.7	80.2	0.82	11.0	0.64			
Pioneer core complex—other areas													
05PC28 (1)	3024	99	120	109	127	75.6	75.8	0.83	9.8	0.05	9.45	43°45.250'	114°07.216'
05PC28 (2)		98	153	166	134	11.8	44.8	0.71	9.1	0.35			
07PC04 (1)	2714	46	55	127	58	23.8	76.1	0.83	9.2	0.34	9.19		
11PC35 (1)	2552	79	37	185	88	11.2	51.4	0.74	8.6	0.47	8.38	43°48.128'	114°10.329'
11PC35 (2)		135	40	330	144	17.0	55.1	0.77	7.7	0.42			
11PC35 (3)		52	26	178	59	7.00	38.6	0.67	8.9	0.47			
Summit Creek Stock													
10SC03 (1)	2696	57	122	1392	86	11.0	103.9	0.87	7.2	0.42	9.51	43°51.557'	114°15.574'
10SC03 (2)		65	166	898	104	8.49	53.3	0.77	5.3	0.29			
10SC03 (3)		43	121	768	71	16.1	74.9	0.82	13.6	0.77			
10SC03 (4)		11	23	127	16	5.06	71.2	0.80	11.9	0.68			
10SC04 (1)	2600	15	41	321	25	11.4	81.6	0.82	6.6	0.38	9.85	43°51.536'	114°15.426'
10SC04 (2)		33	105	566	58	18.9	68.4	0.80	10.2	0.56			
10SC04 (3)		12	22	261	17	13.5	85.4	0.84	12.8	0.75			
10SC07 (1)	2325	23	51	86	35	2.18	43.7	0.69	7.3	0.41	7.70	43°51.040'	114°13.926'
10SC07 (2)		29	39	74	38	6.48	61.5	0.77	8.2	0.47			
10SC07 (3)		22	69	143	39	2.35	40.9	0.68	7.6	0.60			
Garfield Stock													
10GS01 (1)	2863	8.5	42	229	18	9.16	60.2	0.77	15.7	0.88	16.60	43°38.497'	113°54.336'
10GS01 (2)		3.3	16	82	7	8.62	50.1	0.75	17.4	0.95			
10GS01 (3)		1.4	7.0	35	3	4.75	37.5	0.69	16.7	0.98			
10GS04 (1)	2625	69	70	266	86	27.5	54.2	0.77	10.4	0.58	9.62	43°38.327'	113°54.478'
10GS04 (2)		8.3	6.8	33	10	23.9	49.8	0.76	9.6	0.53			
10GS04 (3)		21	16	96	24	21.1	53.3	0.77	8.9	0.49			
Lake Creek Stock													
07PC02 (1)	2558	6.7	11	108	9	5.72	62.5	0.78	29.6	0.45	31.80	43°45.541'	113°54.630'
07PC02 (2)		17	79	369	36	11.6	47.9	0.70	34.0	0.54			
Mackay Stock													
07PC11 (1)	3110	47	200	738	94	73.0	46.6	0.72	44.3	2.36	44.17	43°53.500'	113°42.350'
07PC11 (2)		13	47	344	24	10.4	46.6	0.70	41.4	2.21			
07PC11 (3)		34	156	897	70	41.8	42.0	0.70	46.7	2.45			
Boise Mountains													
11TM01 (1)	2320	5.3	10	62	8	5.64	49.5	0.75	10.6	0.58	10.96	43°36.006'	115°26.838'
11TM01 (2)		8.6	21	135	14	6.47	56.2	0.78	11.1	0.62			
11TM01 (3)		3.9	11	55	6	4.63	46.7	0.72	11.2	0.61			
11BR05 (1)	933	7.0	7.1	76	9	1.50	54.5	0.75	9.5	0.50	9.8	43°36.177'	115°57.404'
11BR05 (2)		40	43	205	50	7.95	50.0	0.72	10.1	0.50			

*Mass-weighted average radius.

[†]Includes 3% α-ejection correction errors.[§]Not shown in Figure 4.F_T—α-ejection correction.

relations exist in the Bell Mountain area, ~17 km to the southwest of Elk Mountain (Fig. 2A). Farther south, late Miocene volcanic rocks overlie Challis volcanics (Fig. 2A). These relationships indicate that these areas have undergone minimal erosional exhumation, while the region ~25 km to the north of Elk Mountain underwent 2–3 km of erosional exhumation.

On the basis of these observations, an estimate of the amount of exhumation along a NW-SE transect is shown by the dashed line in Figure 2C. This line is estimated from (1) AHe data indicating 2–3 km of exhumation in the Pioneer core complex–Summit Creek Stock area since ca. 11 Ma, (2) the preservation of Challis volcanics and Eocene topography suggesting minimal erosion in the Elk Mountain area, and (3) the widespread preservation of Challis volcanics to the north. Further quantification of the northern and southern regions is hindered by the poorly constrained original thickness of the Challis volcanic units.

Thus, the Pioneer core complex–Summit Creek Stock area represents the apex of a Miocene-aged exhumation culmination that spans a distance of 30–35 km (defined by removal of Challis volcanics) in a NW-SE direction (Fig. 2C). This area represents the region of maximum Miocene (and younger) exhumation found within the Pioneer-Boulder Mountains and generally coincides with the highest elevations in the region. Because the base of the Challis volcanics (assumed to have been deposited at roughly similar elevations) occurs at significantly higher elevation in the Pioneer core complex–Summit Creek Stock area, the exhumation culmination can also be regarded as an uplifted culmination.

Between the Pioneer core complex–Summit Creek Stock area and the Hailey Valley to the west, Challis volcanics are locally preserved as erosional remnants (Fig. 2A). In this area, all within the footwall of the W-side-down Sun Valley fault segments, the elevation of the base of the Challis volcanics decreases from >3100–2700 m near the Pioneer core complex to ~2000 m in the immediate footwall (Fig. 2A). The lack of constraints on the original thickness of Challis volcanics precludes a firm estimate of Miocene exhumation. Given that most of the mapped volcanic centers lie to the east or north, it seems likely that the volcanics are thinner than to the east. In the hanging wall of the Sun Valley fault, Challis volcanics are downropped further to elevations ≤~1800 m.

East of the culmination, Challis volcanics are extensively preserved, except in the footwall of the Mackay horst. The 30–34 Ma AHe ages from the northern satellite bodies of the Lake Creek Stock suggest that Miocene exhumation

in this area is less than in the culmination, even in areas where the Challis units have been removed. The highest peaks northwest (Porphyry Peak, elevation ~3050 m) and southeast (Sheep Mountain, elevation ~2940 m) of the Mackay horst (Fig. 2A) are capped by ca. 47 Ma dacites that represent the last stages of Challis volcanic activity in this region (Snider, 1995). In these areas, topographic relief ranges from ~800 m to >1000 m, and the Challis volcanics are preserved throughout most of these elevations. Thus, there appears to be negligible exhumation of the highest elevations, with up to ~1000 m of incision in the adjacent valleys.

The Mackay horst is a topographic high (numerous ridgelines over 3200 m) bounded by NE-trending normal faults (Fig. 2). Our AHe age of ca. 44 Ma from the crest of the Mackay horst indicates motion on the NW-bounding fault in the middle Eocene (discussed earlier herein). The SE-bounding fault displays a more distinct geomorphic expression than the NW-bounding fault, possibly indicating later motion, perhaps in the Miocene. However, the AHe data suggest that Miocene exhumation was limited to depths above the AHe PRZ. An Eocene dike cuts a northeastern strand of this fault (Snider, 1995); however, dikes cutting the main fault have not been reported.

To the southeast of the Pioneer core complex–Summit Creek Stock culmination, where AHe ages from the Garfield Stock suggest ~2 km of post-10 Ma exhumation (see previous), a linear NNW-trending elongate topographic high has an extensive ridgeline above 3000 m (Figs. 2B and 6A). Challis volcanic units have been erosional removed from the peaks and ridges, but they are exposed at lower elevations to the east where they dip 15°–25°E (Fig. 6B; Skipp et al., 2009).

CONTROLS ON EXHUMATION OF THE CULMINATION

In the previous sections, we have quantified the amount of exhumation using AHe ages, and we have attempted to place these localities in a regional context by describing the spatial distribution of exhumation patterns. In this section, we discuss the potential controls on these exhumation patterns. Because it appears likely that uplift was a primary control on erosional exhumation, we discuss two main uplift processes: normal faulting and hotspot-related uplift, with the latter encompassing multiple processes. Ultimately, erosion is the only exhumation process in the region (large-scale Miocene detachment faults do not occur); thus, we also discuss the potential role of surface processes, such as base-level lowering and drainage reorganization.

It is difficult to disentangle the magnitude of the contributions of each of these processes from our data, so our discussion is focused on assessing the relative merits of each.

Role of Faulting

One potential control on exhumation is surface uplift of the footwalls of normal faults, which, coupled with erosion, may produce significant exhumation. However, for the regional Pioneer-Boulder Mountains exhumation culmination, there are no faults with sufficient strike length and throw to produce footwall uplift/exhumation of the 2–3 km magnitude recorded by the AHe ages. For example, along the Hailey Valley to the west, the Sun Valley fault consists of a series of SW-side-down segments (e.g., Dover, 1981; Worl et al., 1991). However, three relationships suggest that this fault is not the primary structure responsible for uplift and exhumation of the Pioneer core complex–Summit Creek Stock culmination: (1) the fault segments are discontinuous and do not have enough throw to produce 2–3 km of exhumation, particularly given that Challis volcanics occur on both sides of the fault, (2) erosional remnants of Challis volcanics preserved in the footwall decrease in elevation toward, rather than away from, the fault (discussed earlier herein; Fig. 2A) as would be expected from an east-tilted footwall, and (3) both the topographic high and the region of maximum exhumation occur 15–25 km away from the fault, whereas footwall topographic highs and drainage divides commonly occur much closer to range-bounding normal faults (e.g., Lost River and Lemhi Ranges to the east, as well as Copper Creek fault; Figs. 1, 2, and 4).

The geologic evidence presented here indicates that the Pioneer core complex–bounding Wildhorse detachment was not active in the Miocene and that samples from both the hanging wall and footwall underwent similar amounts of late Miocene and younger exhumation. The White Mountains fault, which bounds the east side of the Pioneer core complex, has largely been regarded as Eocene (e.g., Silverberg, 1990), but it could have experienced some Miocene motion. However, displacement of this fault decreases to zero at the southwest corner of the Pioneer core complex, where Challis units have been completely removed on both sides of the fault (Fig. 2), indicating that exhumation there was not related to this fault; it also cannot account for exhumation in the western Pioneer core complex and Summit Creek Stock areas.

The Copper Creek fault to the southeast of the Pioneer core complex is a W-side-down normal fault with a throw of ~2–4 km based on cross-sectional reconstruction (Figs. 2A and 6).

Footwall uplift is indicated by a fault-parallel elongate topographic high and drainage divide ~4–5 km east of the fault trace (Fig. 6). Our AHe ages suggest that motion on this fault began at or after ca. 9.6 Ma (see earlier herein). Both the displacement and geomorphic expression of the Copper Creek fault disappear a few kilometers south of the White Mountains fault that bounds the southeast side of the Pioneer core complex.

A N-trending, W-side-down fault (referred to here as the Airport fault) also occurs on the north side of the Pioneer core complex (Fig. 2; Dover, 1981). This fault loses displacement to the south, where it does not significantly offset the Pioneer core complex detachment. Due to the lack of any apatite-bearing lithologies in the footwall of the Airport fault, we have not constrained its age. However, given the similar polarity and along-strike location to the Copper Creek fault, it may have formed at roughly the same time.

Motion on the Copper Creek fault (and perhaps the Airport fault) at ca. ≤ 9.6 Ma may have overlapped in age with exhumation in the Pioneer core complex, which began around 11 Ma and continued to < 8 Ma. However, since neither fault continues into the Pioneer core complex, these faults did not contribute to uplift and exhumation of the Pioneer core complex. Thus, extensional faulting occurred within the broad regional culmination, but the discontinuous fault segments did not link to form a through-going regional fault system.

In summary, although the second-order detailed geometry of the exhumed area may be affected by small-displacement faults, these faults do not have large enough throws or lengths or discernible geometric relationships with the exhumation patterns and distribution of topography to have been the primary control on uplift and exhumation of the regional-scale culmination. Some of the faults have produced more localized regions of uplift/exhumation that are superimposed on this regional culmination, forming either during or after culmination development.

Role of Hotspot-Related Processes

Because faulting cannot account for the primary geometry and magnitude of the Pioneer-Boulder Mountains regional exhumation patterns, we must explore other potential controls. The timing and location of exhumation of the Pioneer-Boulder Mountains culmination suggest a relationship between uplift/exhumation and formation of the Picabo volcanic field beginning around 10.3 Ma. We envision two potential surface uplift processes that link exhumation to hotspot phenomena: (1) thermal-dynamic uplift and (2) flexural uplift. Surface

uplift from these mechanisms would potentially lead to an increase in erosional exhumation rate.

Since thermal-dynamic uplift is driven by mantle heat flow and flexural uplift is driven by crustal emplacement of mantle-derived mafic magmas, these two general processes are inherent in any mantle-driven model (e.g., plume or slab-controlled edge convection) for Snake River Plain–Yellowstone volcanism. Thus, the operation of one or both of these processes does not distinguish between plume and nonplume models.

Thermal-Dynamic Uplift

Several workers have suggested that the elevation anomaly centered around the Yellowstone region today is the result of a thermal-dynamic uplift associated with the rising tail of a mantle plume (Suppe et al., 1975; Pierce and Morgan, 1992; Smith and Braile, 1993; Waschbusch and McNutt, 1994). The uplifted area is generally regarded to be ~0.5–1.0 km in height and 350–400 km across (e.g., Pierce and Morgan, 1992, 2009; Smith et al., 2009). Assuming a similar topographic swell was centered on the Picabo volcanic field at around 10.3 Ma, the elevated topography and the Pioneer-Boulder Mountains exhumation culmination would lie within the inner part of such a swell. Thus, the increase in exhumation rate shown by our AHe data in the Pioneer-Boulder Mountains could have been facilitated by thermal-dynamic uplift over the plume tail beginning prior to hotspot arrival at ca. 10.3 Ma. Although thermal-dynamic uplift is probably responsible for a component of the large-scale uplift of the Yellowstone crescent of high terrain, it is difficult to ascribe the 2–3 km differences in exhumation over distances of as little as 25 km and the overall exhumation geometry shown in Figure 2 solely to a broad thermal-dynamic uplift.

Flexural Uplift and Eastern Snake River Plain Downflexing

Another potential hotspot-related uplift process is loading of the eastern Snake River Plain by mantle-derived mafic magmas, producing elastic flexural uplift along its flanks. Loading was modeled by McQuarrie and Rodgers (1998) to fit the downflexed topography at the southern ends of the Lost River, Lemhi, and Beaverhead Mountains to the east of the Pioneer-Boulder Mountains. Although loading of the flexed elastic plate would be expected to produce uplift (peripheral bulge) north of the eastern Snake River Plain, the models of McQuarrie and Rodgers did not extend far enough northward to address potential uplift. Furthermore, their modeling was focused in the fault-bounded Lost River, Lemhi, and Beaverhead Ranges, where

it is difficult to discriminate between variations in footwall uplift along the fault strikes and flexural bulging associated with eastern Snake River Plain loading.

Downflexing of the southern end of the Pioneer Mountains is recorded by geologic relations and dating of Miocene tuffs in the Lake Hills area mapped by Michalek (2009), who showed that southward tilting occurred between emplacement of tuffs dated at 9.2 ± 0.2 and 8.4 ± 0.5 Ma (with additional tilting after ca. 4.2 Ma). On the basis of northward thinning of the ca. 9.2 Ma tuff and greater tilting of Mesozoic fold hinges in the region (McQuarrie and Rodgers, 1998) compared with the 9.2 Ma tuff, Michalek (2009) inferred that southward tilting began prior to 9.2 Ma. This tilting was likely the result of loading of the eastern Snake River Plain by emplacement of mantle-derived mafic magmas into the middle crust (McQuarrie and Rodgers, 1998; Rodgers et al., 2002; Michalek, 2009), which produced a network of sills estimated to be 10–12 km thick (Braile et al., 1982; Sparlin et al., 1982; Peng and Humphreys, 1998; DeNosaquo et al., 2009). Through melting of the middle crust, these mafic magmas were responsible for silicic volcanism, including the 10.3–10.1 Ma Tuff of Arbon valley (e.g., Kellogg et al., 1994) and 8.2 Ma rhyolite in the INEL-1 well (Fig. 1; McCurry and Rodgers, 2009) erupted from the inferred Picabo volcanic field (Pierce and Morgan, 1992).

Thus, loading of the eastern Snake River Plain by mafic magmas and associated southward downflexing of the southern Pioneer Mountains occurred during the time of high exhumation rates to the north, as recorded by our AHe ages. This synchronicity, combined with the location and cross-sectional geometry of the exhumation culmination, suggests that flexural uplift may have played a significant role in exhumation in the Pioneer core complex–Summit Creek Stock area. We further suggest that the limited exhumation south of the Pioneer core complex and the preservation of Eocene topography (e.g., Elk Mountain topographic high) were the result of lesser (or lack of) uplift to the south compared with the Pioneer core complex–Summit Creek Stock area.

A rough estimate of the amount of potential flexural uplift can be derived from basic elastic plate analysis and the modeling of McQuarrie and Rodgers (1998). Turcotte and Schubert (2002) showed that the uplift of the flexural bulge is equal to $0.067W_0$ (where W_0 is the maximum deflection beneath the line load) for a broken plate, which may be a valid assumption given that the eastern Snake River Plain lithosphere may have been weakened by extensive magmatism. Modeling by McQuarrie and

Rodgers (1998), using a rectangular load on the eastern Snake River Plain, yielded a deflection of ~9 km beneath the eastern Snake River Plain. Using this value of W_0 yields ~600 m of expected flexural uplift, while an unbroken plate yields a lower value of ~390 m.

While flexural uplift appears consistent with the uplift/exhumation pattern in a two-dimensional NNW-SSE cross section (Fig. 2C), the Pioneer-Boulder Mountains culmination does not continue in an obvious way to the NE, as would be expected from a line load applied to the eastern Snake River Plain. One possible explanation is that to the east, surface uplift is offset from the Pioneer core complex–Summit Creek Stock area, occurring in the Mackay horst and the ridge of high elevation along strike to the southwest (Fig. 2B). This uplift may have been accommodated by the normal fault on the SE side of the horst, but it did not produce more than ~1.0–1.5 km of exhumation (on the basis of AHe data discussed earlier). Motion on this, and other NE-trending faults may be related to bending stresses caused by loading of the eastern Snake River Plain (Zentner, 1989; Rodgers et al., 2002).

The lack of an obvious exhumation culmination to the northeast may also be explained by NE tilting of the eastern side of the Pioneer-Boulder Mountains from SW-side-down motion on the Lost River fault (Fig. 1). Eastward tilting of the hanging-wall topographic surface could have limited the rate of erosion on this side of the range and also obscured an earlier, more extensive linear flexural uplift that paralleled the eastern Snake River Plain.

Role of Surface Processes

Ultimately, the exhumation recorded by our AHe thermochronology and by the absence of Challis volcanics was caused by erosion. Although erosion may have been largely focused in areas undergoing surface uplift, erosion rates may have been controlled by additional factors, such as lowering of base level and drainage reorganization.

During postorogenic Eocene extension, rivers drained eastward and southeastward from the study area into the foreland region (e.g., Janecke et al., 2000; Chetel et al., 2011). Downflexing and surface subsidence of the eastern Snake River Plain (McQuarrie and Rodgers, 1998) would have caused reorganization of the regional drainage pattern and integration of the Big Wood and Big Lost Rivers, which currently flow south into the eastern Snake River Plain (Figs. 1 and 2). Subsidence and drainage integration would produce a regional drop in base level, and, since most of the Pioneer-Boulder

Mountains culmination is drained by these two fluvial systems, erosional potential in the culmination may have been significantly increased in the Miocene. Hodges et al. (2009) showed that ca. 695 Ma detrital zircons derived from the Pioneer core complex were deposited in the central Snake River Plain by the earliest Pliocene via the Big Lost River. Similarly, detrital zircons in drill core from the Kimama well (Fig. 1) indicate that the Big Wood River flowed south into the eastern Snake River Plain by ca. 6 Ma (P. Link, 2013, oral commun.; Potter et al., 2013). Both of these areas provide only a minimum age for drainage integration. It is possible, and perhaps likely, that these drainages flowed south by 10–8 Ma, given the fact that downwarping has been demonstrated for much of the length of the eastern Snake River Plain at this time (Rodgers et al., 2002).

The evolving pattern of drainage integration may also explain the lack of Challis removal in the Boulder Mountains to the north of the North Fork Big Lost River, where thick deposits of Challis volcanics are extensively preserved (Fig. 2A) despite the high elevations (Figs. 2B and 2C). Except for the southern edge, this area is drained by the north- and northeast-flowing Salmon River tributaries and therefore has not felt the effects of base-level fall in the eastern Snake River Plain, since these drainages are not integrated with the Snake River. It is also possible that uplift propagated northward and is therefore more recent in the Boulder Mountains (consistent with loading and flexural uplift). In summary, the exhumation culmination in the Pioneer-Boulder Mountains generally coincides with an area that drains into the Snake River Plain, and this drainage pattern reflects topographic changes that occurred in the late Miocene.

REGIONAL EXTENT OF EXHUMATION AT 11–8 MA

Published studies and our new data indicate that volcanism, extension, and exhumation were all widespread during the approximate time period over which the Pioneer-Boulder Mountains were exhumed (i.e., 11–8 Ma). Wood and Clemens (2002) have shown that rhyolitic volcanism occurred along the southwest and northeast margins of the western Snake River Plain at ca. 12–9 Ma, while Bonnicksen et al. (2008) documented widespread silicic volcanism at ca. 12–10 Ma across a broad area of the central Snake River Plain corresponding to the areas of the Bruneau-Jarbridge and Twin Falls volcanic centers. This time period included eruption of the 10.3–10.1 Ma Tuff of Arbon Valley (e.g., Kellogg et al., 1994; Anders et al., 2009) from the inferred Picabo volcanic center, as well as

basaltic volcanism as far east as the southern end of the Lemhi Range (Fig. 1A).

Extensional faulting was widespread during this time period as well. Wood and Clemens (2002) interpreted the western Snake River Plain as a graben with marginal faulting occurring largely at ca. 11–9 Ma. Other faulted regions that were active at this time include the Rogerson graben (Andrews et al., 2008) and the Raft River detachment (Wells et al., 2000) south of the Snake River Plain (Fig. 1). Our initial AHe ages and published apatite fission-track ages (Sweetkind and Blackwell, 1989) discussed herein also indicate that significant exhumation was occurring in the Boise Mountains contemporaneously with exhumation occurring in the Pioneer-Boulder Mountains.

Thus, exhumation and volcanism at ca. 12–9 Ma were widespread, and there is a spatial coincidence of the two, suggesting a causal link between exhumation and hotspot processes. Several authors have suggested that this anomalous period coincides with the time of transition from the plume head to plume tail, which may also have involved tilting of the plume (Shervais and Hanan, 2008; Pierce and Morgan, 2009). However, the nature and position of the plume during this time are poorly understood, and a number of direct and indirect hotspot controls may have contributed to the Miocene exhumation recorded by low-temperature thermochronology, including: dynamic hotspot-related uplift, flexure, and base-level lowering by extensional faulting in the western Snake River Plain.

SUMMARY AND CONCLUDING REMARKS

Our new AHe data show that >2–3 km of exhumation has occurred in the Pioneer-Boulder Mountains since ca. 11 Ma, with rates estimated at ~0.3 mm/yr. This late Miocene exhumation appears to have followed a period of >30 m.y. during which very little regional-scale exhumation occurred. The locus of exhumation and the synchronicity between rapid exhumation and the arrival of the Yellowstone hotspot suggest that uplift and exhumation are related to hotspot processes. Furthermore, the general lack of correlation between exhumed regions and Miocene normal faults suggests that extensional faulting is not the primary control on topography and exhumation in the Pioneer-Boulder Mountains, as appears to be the case for linear fault-bounded ranges to the east. Extensional faulting in the Pioneer-Boulder Mountains instead appears to be superimposed on a broad culmination.

From our discussion, it is clear that the exhumation and topographic patterns in the study area are a complicated result of multiple

hotspot-related uplift processes combined with faulting and surface processes. We have presented a brief discussion of the potential relevance of these processes. At present, however, disentanglement of the contributing effects of each is difficult. Some of the ambiguity stems from the lack of full spatial distribution of exhumation introduced by the uncertainties in the pre-11 Ma thicknesses of Challis volcanics and by the limited distribution of apatite-bearing lithologies for AHe thermochronology. Furthermore, the lack of constraints on the ages of the Lost River, Sun Valley, and Hailey fault systems provides additional uncertainty in attempting to discern the hotspot effects of uplift/exhumation. Lastly, the lack of complete coupling between surface uplift and exhumation hinders a full quantitative assessment of the spatial distribution of Miocene vertical movements. Our ongoing thermochronologic study expanded over a broader region will address some of these issues by providing a clearer picture of the regional spatial-temporal patterns of exhumation along the length of the Snake River Plain–Yellowstone system.

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